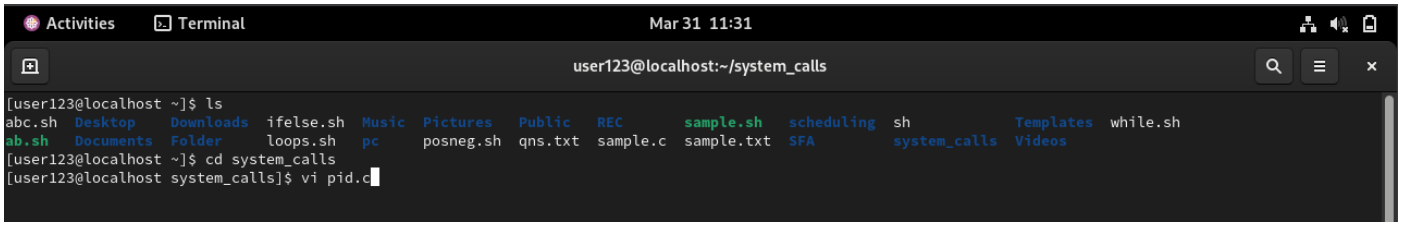
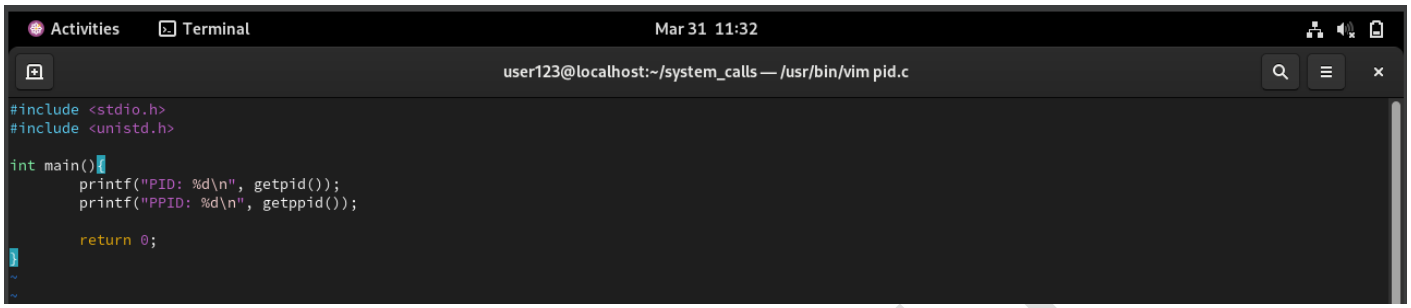
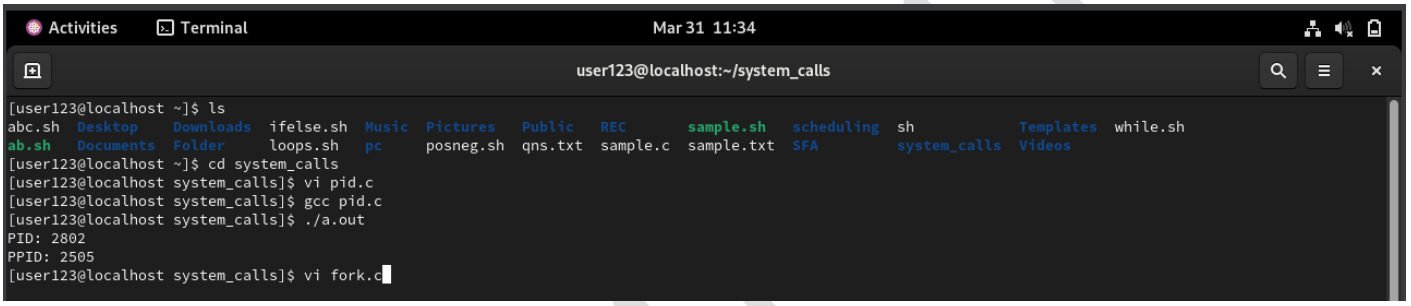
SYED FAIYAZ AHAMED
241401113

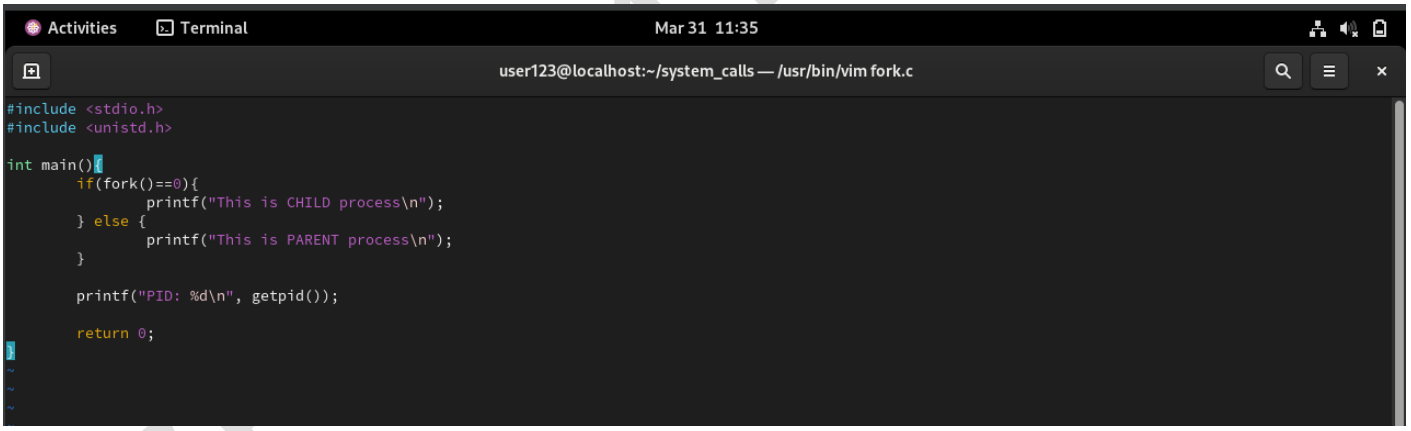
OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

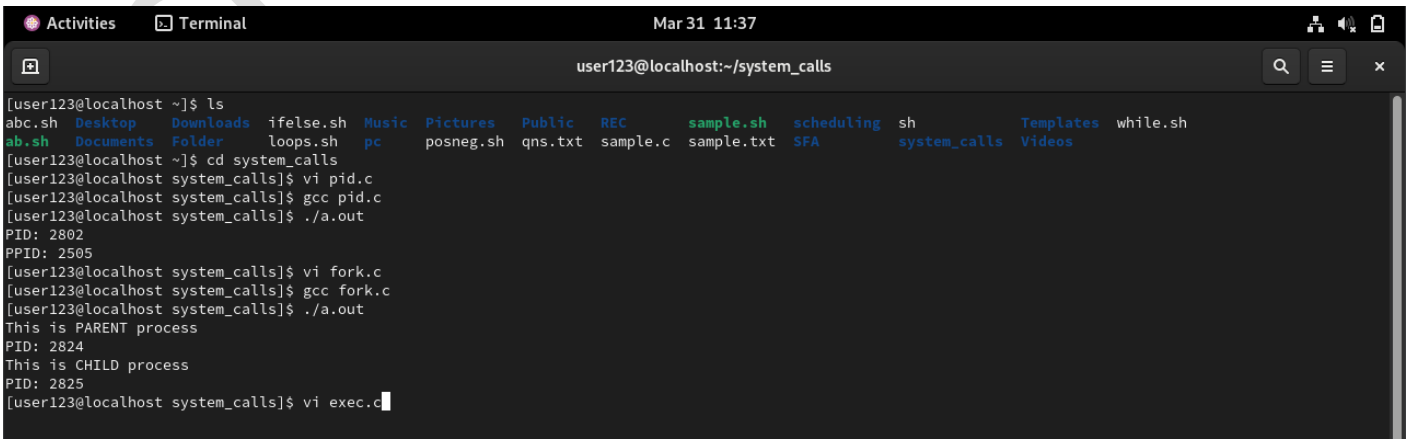
EXP – 04: SYSTEM CALLS

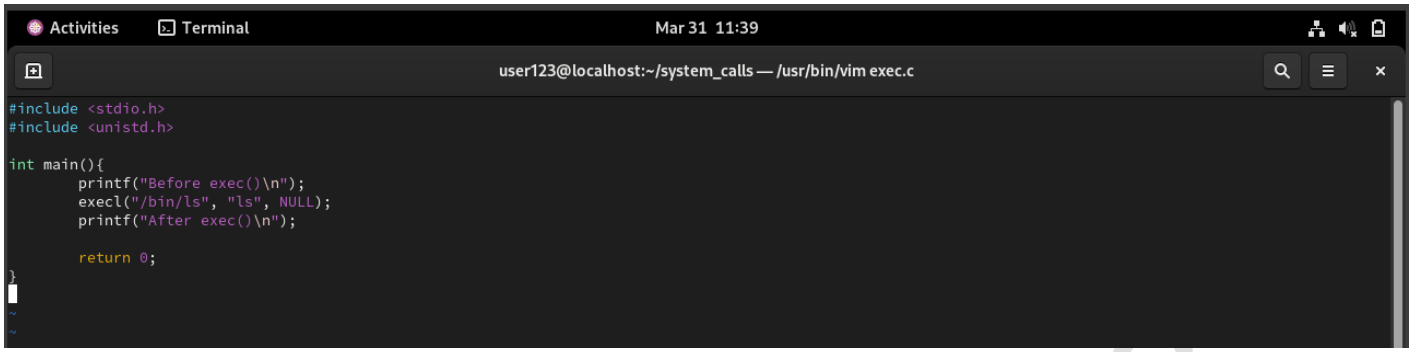


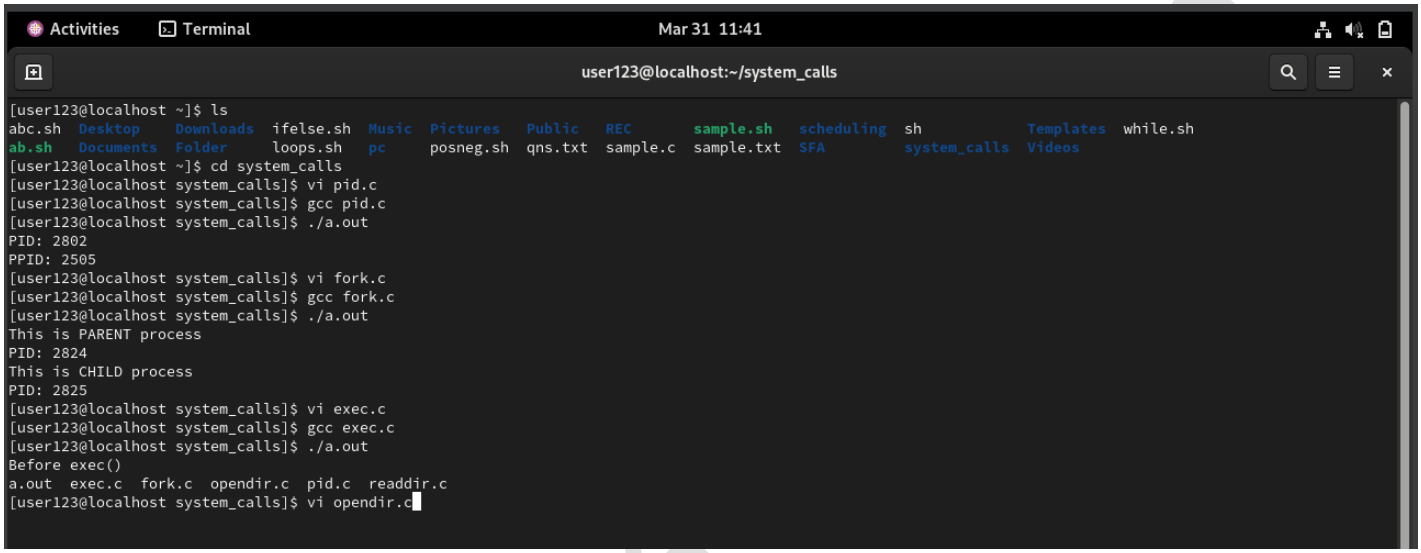




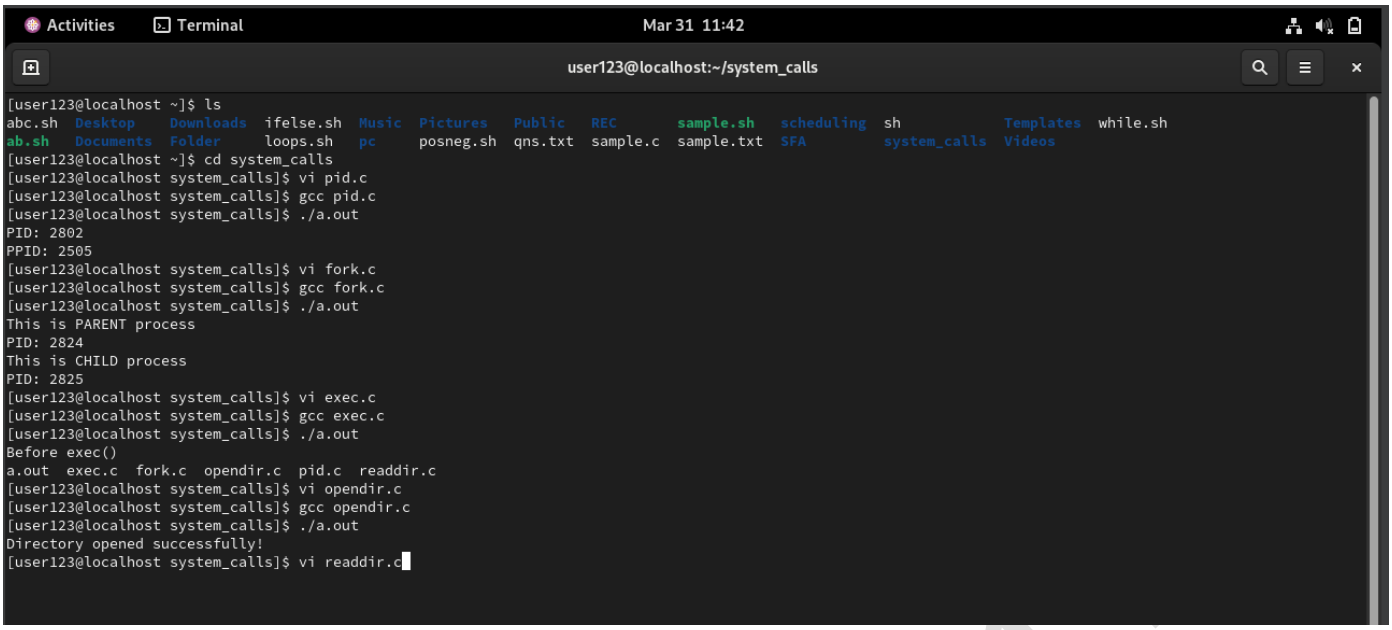




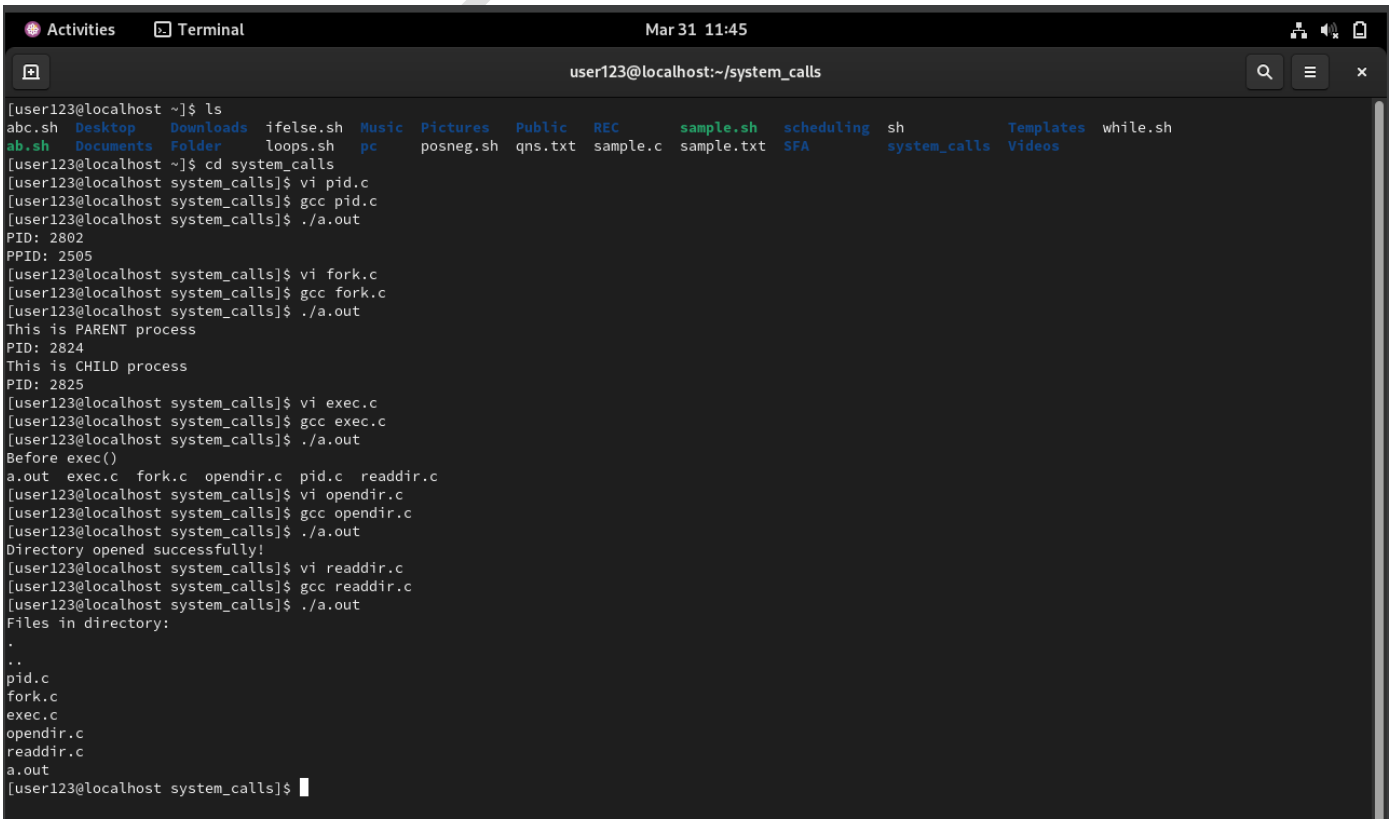

Activities
Terminal
Mar 31 11:39
user123@localhost:~/system_calls — /usr/bin/vim exec.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Before exec()\n");
execl("/bin/ls", "ls", NULL);
printf("After exec()\n");
return 0;
}


Activities
Terminal
Mar 31 11:41
user123@localhost:~/system_calls
[user123@localhost ~]$ ls
abc.sh Desktop Downloads ifelse.sh Music Pictures Public REC sample.sh scheduling sh Templates while.sh
ab.sh Documents Folder loops.sh pc posneg.sh qns.txt sample.c sample.txt SFA system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out exec.c fork.c opendir.c pid.c readdir.c
[user123@localhost system_calls]$ vi opendir.c


Activities
Terminal
Mar 31 11:41
user123@localhost:~/system_calls — /usr/bin/vim opendir.c
#include <stdio.h>
#include <dirent.h>
int main(){
DIR *dir;
dir = opendir(".");
if(dir == NULL){
printf("Failed to open directory\n");
return 1;
}
printf("Directory opened successfully!\n");
closedir(dir);
return 0;
}


Activities
Terminal
Mar 31 11:42
user123@localhost:~/system_calls
[user123@localhost ~]$ ls
abc.sh Desktop Downloads ifelse.sh Music Pictures Public REC sample.sh scheduling sh Templates while.sh
ab.sh Documents Folder loops.sh pc posneg.sh qns.txt sample.c sample.txt SFA system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out exec.c fork.c opendir.c pid.c readdir.c
[user123@localhost system_calls]$ vi opendir.c
[user123@localhost system_calls]$ gcc opendir.c
[user123@localhost system_calls]$ ./a.out
Directory opened successfully!
[user123@localhost system_calls]$ vi readdir.c


Activities
Terminal
Mar 31 11:44
user123@localhost:~/system_calls — /usr/bin/vim readdir.c
#include <stdio.h>
#include <dirent.h>
int main(){
DIR *dir;
struct dirent *entry;
dir = opendir(".");
if(dir == NULL){
printf("Cannot open directory\n");
return 1;
}
printf("Files in directory:\n");
while((entry=readdir(dir)) != NULL){
printf("%s\n", entry->d_name);
}
closedir(dir);
return 0;
}


Activities
Terminal
Mar 31 11:45
user123@localhost:~/system_calls
[user123@localhost ~]$ ls
abc.sh Desktop Downloads ifelse.sh Music Pictures Public REC sample.sh scheduling sh Templates while.sh
ab.sh Documents Folder loops.sh pc posneg.sh qns.txt sample.c sample.txt SFA system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out exec.c fork.c opendir.c pid.c readdir.c
[user123@localhost system_calls]$ vi opendir.c
[user123@localhost system_calls]$ gcc opendir.c
[user123@localhost system_calls]$ ./a.out
Directory opened successfully!
[user123@localhost system_calls]$ vi readdir.c
[user123@localhost system_calls]$ gcc readdir.c
[user123@localhost system_calls]$ ./a.out
Files in directory:
.
..
pid.c
fork.c
exec.c
opendir.c
readdir.c
a.out
[user123@localhost system_calls]$